NATIONAL RESERVE BANK OF TONGA



Nowcasting Tongan Output

September 2026

Outline

1. Tonga Data Management
2. Inflation Nowcast
3. GDP Nowcast
4. Import Nowcast
5. Macroeconomic Diagnostics
6. QPM Outputs
7. Key Lessons

1. Tonga Data Management: Development

Eviews Name	cpi_u	cpiim_u	cpi_dom	cpif_u
Stocks/flow	stock	stock	stock	stock
Frequency	Monthly	Monthly	Monthly	Monthly
Conversion method	last	last	last	last
Interpolation method				
Position of observation				
Title	Consumer Price Index_unseasonalized	Consumer Price Index_imports	Consumer Price Index_local	Consumer Price Index_food
Data source/Link	NRBT/Economics Dept.	NRBT/Economics Dept.	NRBT/Economics Dept.	NRBT/Economics Dept.
Database name	Price_new	Price_new	Price_new	Price_new
Unit	Index	Index	Index	Index
Sector	Real	Real	Real	Real
Last revision date				
Economist	Leonard Fatukala	Leonard Fatukala	Leonard Fatukala	Leonard Fatukala
Last update date	30-Jul-26	30-Jul-26	30-Jul-26	30-Jul-26
Period	CPI	Import	Domt	Food
1/31/2025	133.4	118.0	149.8	144.0
2/28/2025	129.8	118.3	142.0	134.3
3/31/2025	129.1	118.4	140.3	131.8
4/30/2025	129.4	118.8	140.7	131.1
5/31/2025	130.3	119.5	141.8	132.5
6/30/2025	133.8	119.5	148.9	140.3
7/31/2025	131.2	119.7	143.4	135.6
8/31/2025	130.2	120.3	140.6	132.9
9/30/2025	130.6	120.5	141.3	134.3
10/31/2025	130.5	120.4	141.2	133.2
11/30/2025	133.8	126.0	142.2	134.5
12/31/2025	136.9	127.3	147.1	140.9
1/31/2026	136.1	125.1	146.7	138.5
2/28/2026	133.0	123.8	142.7	132.5
3/31/2026	134.9	126.6	143.8	133.5
4/30/2026	136.5	129.3	144.2	133.9
5/31/2026	140.7	133.5	148.2	135.4
6/30/2026	143.3	135.3	151.8	139.2

Read me | Daily (EV) | Monthly (EV) | Quarterly (EV) | Annual (EV) | BOP_annual | +

- **Tonga Data Management:**

  Relevant economic data for Nowcasting and the Quarterly Projection Models are compiled into a single Excel file, by frequency.

  Specify key attributes for every data series

  An EViews script is used to update this data into EViews prior to running the Nowcast and QPM processes.

  While the master custodian ensures that all data are consistently maintained and kept up to date, each economist is responsible for updating their respective data sets.

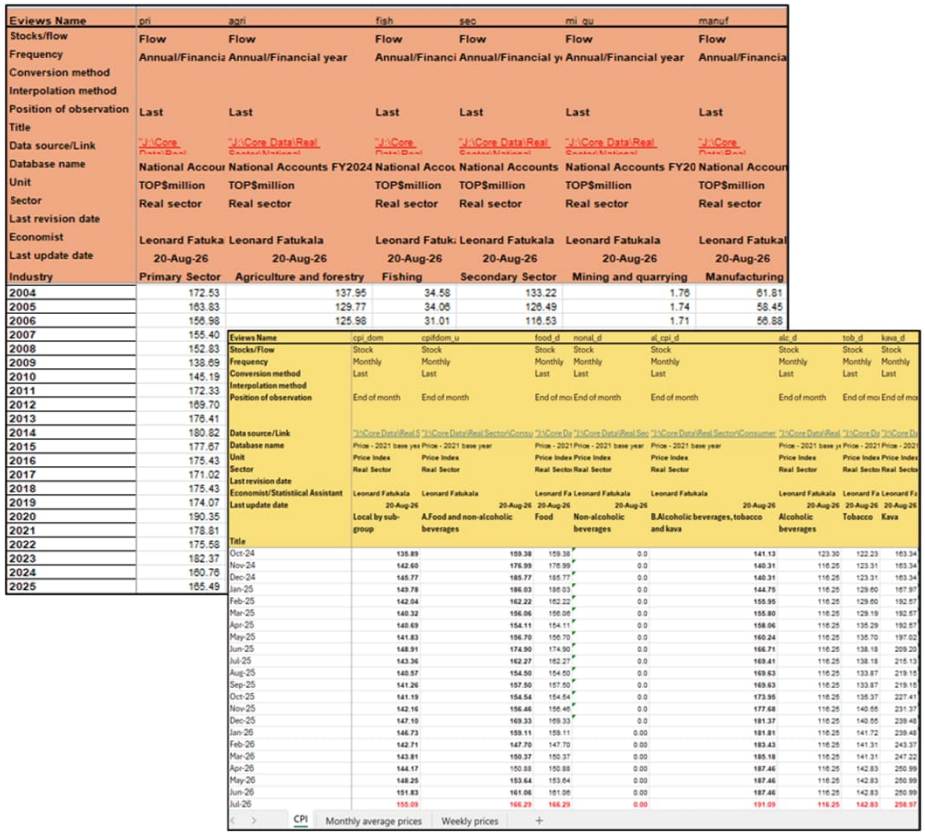
Tonga Data Management: Challenges

Eviews Name	cpi_u	cpiim_u	cpi_dom	cpif_u
Stocks/flow	stock	stock	stock	stock
Frequency	Monthly	Monthly	Monthly	Monthly
Conversion method	last	last	last	last
Interpolation method				
Position of observation				
Title	Consumer Price Index_unseasonalized	Consumer Price Index_imports	Consumer Price Index_local	Consumer Price Index_food
Data source/Link	NRBT/Economics Dept.	NRBT/Economics Dept.	NRBT/Economics Dept.	NRBT/Economics Dept.
Database name	Price_new	Price_new	Price_new	Price_new
Unit	Index	Index	Index	Index
Sector	Real	Real	Real	Real
Last revision date				
Economist	Leonard Fatukala	Leonard Fatukala	Leonard Fatukala	Leonard Fatukala
Last update date	30-Jul-26	30-Jul-26	30-Jul-26	30-Jul-26
Period	CPI	Import	Domt	Food
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5/31/2025	130.3	119.5	141.8	132.5
6/30/2025	133.8	119.5	148.9	140.3
7/31/2025	131.2	119.7	143.4	135.6
8/31/2025	130.2	120.3	140.6	132.9
9/30/2025	130.6	120.5	141.3	134.3
10/31/2025	130.5	120.4	141.2	133.2
11/30/2025	133.8	126.0	142.2	134.5
12/31/2025	136.9	127.3	147.1	140.9
1/31/2026	136.1	125.1	146.7	138.5
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3/31/2026	134.9	126.6	143.8	133.5
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6/30/2026	143.3	135.3	151.8	139.2

Read me | Daily (EV) | Monthly (EV) | Quarterly (EV) | Annual (EV) | BOP_annual | +

- One challenge is that the TAs recommended importing all the data into this file, but Excel storage is limited.
- Not all data are in this file, hence data gaps can create extra work in finding missing information and keeping sources consistent.
- A further challenge is that whenever new columns are added, the column command in the EViews script must also be updated.
  - ✓ Operators may occasionally overlook this step, which can lead to confusion and force them to go back and forth checking whether the script is correctly picking up the data.

Tonga Data Management: Next steps



Eviews Name	pri	agri	fish	sec	mi_qu	manuf
Stocks/flow	Flow	Flow	Flow	Flow	Flow	Flow
Frequency	Annual/Financi	Annual/Financial year	Annual/Financi	Annual/Financial y	Annual/Financial year	Annual/Financia
Conversion method						
Interpolation method						
Position of observation	Last	Last	Last	Last	Last	Last
Title						
Data source/Link						
Database name	National Accour	National Accounts FY2024	National Accou	National Accounts	National Accounts FY20	National Accoun
Unit	TOP$million	TOP$million	TOP$million	TOP$million	TOP$million	TOP$million
Sector	Real sector	Real sector	Real sector	Real sector	Real sector	Real sector
Last revision date						
Economist	Leonard Fatuka	Leonard Fatukala	Leonard Fatuk	Leonard Fatukala	Leonard Fatukala	Leonard Fatukal
Last update date	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26
Industry	Primary Sector	Agriculture and forestry	Fishing	Secondary Sector	Mining and quarrying	Manufacturing
2004	172.53	137.95	34.58	133.22	1.76	61.81
2005	163.83	129.77	34.06	126.49	1.74	58.45
2006	156.98	125.98	31.01	116.53	1.71	56.88
2007	155.40					
2008	152.83					
2009	138.69					
2010	145.19					
2011	172.33					
2012	169.70					
2013	176.41					
2014	180.82					
2015	177.67					
2016	175.43					
2017	171.02					
2018	175.43					
2019	174.07					
2020	190.35					
2021	178.81					
2022	175.58					
2023	182.37					
2024	180.76					
2025	185.49					

Eviews Name	cpi_dom	cpiifdom_u	food_d	nonal_d	al_cpi_d	alc_d	tob_d	kava_d
Stocks/Flow	Stock	Stock	Stock	Stock	Stock	Stock	Stock	Stock
Frequency	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly	Monthly
Conversion method	Last	Last	Last	Last	Last	Last	Last	Last
Interpolation method								
Position of observation	End of month	End of month	End of mo	End of month	End of month	End of month	End of mo	End of mo
Data source/Link								
Database name	Price - 2021 base ye	Price - 2021 base year	Price - 2021	Price - 2021 base year	Price - 2021 base year	Price - 2021 base ye	Price - 2021	Price - 2021
Unit	Price Index	Price Index	Price Index	Price Index	Price Index	Price Index	Price Index	Price Index
Sector	Real Sector	Real Sector	Real Sector	Real Sector	Real Sector	Real Sector	Real Sector	Real Sector
Last revision date								
Economist/Statistical Assistant	Leonard Fatukala	Leonard Fatukala	Leonard Fa	Leonard Fatukala	Leonard Fatukala	Leonard Fatukala	Leonard Fa	Leonard Fa
Last update date	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26	20-Aug-26
Title	Local by sub-group	A.Food and non-alcoholic beverages	Food	Non-alcoholic beverages	B.Alcoholic beverages, tobacco and kava	Alcoholic beverages	Tobacco	Kava
Oct-24	135.89	159.38	159.38	0.0	141.13	123.30	122.23	163.34
Nov-24	142.60	176.99	176.99	0.0	140.31	116.25	123.31	163.34
Dec-24	145.77	185.77	185.77	0.0	140.31	116.25	123.31	163.34
Jan-25	149.78	186.03	186.03	0.0	144.75	116.25	129.60	167.97
Feb-25	142.04	162.22	162.22	0.0	155.95	116.25	129.60	192.57
Mar-25	140.32	156.06	156.06	0.0	155.80	116.25	129.19	192.57
Apr-25	140.69	154.11	154.11	0.0	158.06	116.25	135.29	192.57
May-25	141.83	156.70	156.70	0.0	160.24	116.25	135.70	197.02
Jun-25	148.91	174.90	174.90	0.0	166.71	116.25	138.18	209.20
Jul-25	143.36	162.27	162.27	0.0	169.41	116.25	138.18	216.13
Aug-25	140.57	154.50	154.50	0.0	169.63	116.25	133.87	219.15
Sep-25	141.26	157.50	157.50	0.0	169.63	116.25	133.87	219.15
Oct-25	141.19	154.54	154.54	0.0	173.95	116.25	135.37	227.41
Nov-25	142.16	156.46	156.46	0.0	177.68	116.25	140.55	231.37
Dec-25	147.10	169.33	169.33	0.0	181.37	116.25	140.55	239.48
Jan-26	146.73	159.11	159.11	0.00	181.81	116.25	141.72	239.48
Feb-26	142.71	147.70	147.70	0.00	183.43	116.25	141.31	243.37
Mar-26	143.81	150.37	150.37	0.00	185.18	116.25	141.31	247.22
Apr-26	144.17	150.88	150.88	0.00	187.46	116.25	142.83	250.99
May-26	148.25	153.64	153.64	0.00	187.46	116.25	142.83	250.99
Jun-26	151.83	161.06	161.06	0.00	187.46	116.25	142.83	250.99
Jul-26	155.09	166.29	166.29	0.00	191.09	116.25	142.83	258.97

CPI | Monthly average prices | Weekly prices

- Establish a sector-specific centralized database for GDP and Inflation (currently in progress), designed for efficient import into EViews.
- Develop similar databases across other sectors to ensure consistency and accessibility.
- Create an import script to consolidate all sector-specific databases into a single EViews work file.
- Use consistent naming conventions (EViews terms) across all centralized databases.

2. Inflation Nowcast

Model 1:

Dependent Variable: INFL_AD
Method: Least Squares
Date: 09/10/26 Time: 10:49
Sample (adjusted): 2016M01 2026M07
Included observations: 127 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.469690	0.219201	2.142735	0.0341
INFL_AD(-1)	0.899277	0.034845	25.80815	0.0000
OIL_T_INFL(-1)	0.010174	0.004515	2.253354	0.0260
USD_I_A	-0.016019	0.051896	-0.308670	0.7581
NZ_DAIRY_INF(-1)	0.001295	0.016153	0.080142	0.9363

R-squared	0.879741	Mean dependent var	5.015350
Adjusted R-squared	0.875798	S.D. dependent var	4.630548
S.E. of regression	1.631912	Akaike info criterion	3.855956
Sum squared resid	324.9027	Schwarz criterion	3.967932
Log likelihood	-239.8532	Hannan-Quinn criter.	3.901450
F-statistic	223.1190	Durbin-Watson stat	1.959024
Prob(F-statistic)	0.000000		

Model 2:

Dependent Variable: INFL_AD
Method: Least Squares
Date: 09/10/26 Time: 11:11
Sample (adjusted): 2019M01 2026M07
Included observations: 91 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.239723	0.228467	1.049269	0.2970
INFL_AD(-1)	0.684138	0.060299	11.34572	0.0000
OIL_T_INFL	0.015429	0.004452	3.465178	0.0008
USD_I_A	-0.038119	0.079433	-0.479883	0.6325
NZ_DAIRY_INF(-1)	0.105266	0.038713	2.719108	0.0079
FOOD_DOM_INF	0.077060	0.017656	4.364644	0.0000

R-squared	0.921821	Mean dependent var	4.504918
Adjusted R-squared	0.917223	S.D. dependent var	4.901877
S.E. of regression	1.410323	Akaike info criterion	3.589174
Sum squared resid	169.0659	Schwarz criterion	3.754725
Log likelihood	-157.3074	Hannan-Quinn criter.	3.655964
F-statistic	200.4504	Durbin-Watson stat	1.206576
Prob(F-statistic)	0.000000		

- Model 1, built with the IMF technical assistants, uses lagged inflation, lagged oil inflation, NEER (TOP/USD), and NZ dairy inflation.
- Model 2 adds domestic food inflation and current oil inflation. It projects a rapid decline in inflation, conflicting with Model 1 and the Excel-based approach.

Inflation Excel-based Forecast

Consumer Price Index Base Period: September 2021 = 100		Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Apr-27	May-27
All items		130.6	130.5	133.8	136.9	136.1	133.0	134.9	136.5	140.7	143.3	143.4	142.2	143.3	143.9	145.6	147.5	147.1	146.8	146.5	147.7	148.0
J. Education	1.2	106.0	106.0	106.0	106.0	110.2	110.2	110.2	110.24	110.2	110.2	110.2	110.2	110.2	110.2	110.2	110.2	111.2	111.2	111.2	111.2	111.2
Early childhood and primary education	0.1	94.7	94.7	94.7	94.7	101.2	101.2	101.2	101.22	101.2	101.2	101.2	101.2	101.2	101.2	101.2	101.2	101.6	101.6	101.6	101.6	101.6
Secondary education	0.5	117.4	117.4	117.4	117.4	125.7	125.7	125.7	125.72	125.7	125.7	125.7	125.7	125.7	125.7	125.7	125.7	128.0	128.0	128.0	128.0	128.0
Tertiary education	0.6	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.11	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1	99.1
K. Restaurants and hotels	15.0	136.8	136.8	136.8	136.8	147.9	147.9	147.9	147.86	147.9	149.8	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9
Food and beverage serving services	15.0	136.8	136.8	136.8	136.8	147.9	147.9	147.9	147.86	147.9	149.8	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9	156.9
L. Miscellaneous goods and services	5.4	132.7	132.8	140.1	139.0	136.8	133.3	133.4	134.34	135.1	135.1	134.7	135.3	135.9	136.4	137.0	137.6	138.2	138.2	138.2	138.2	138.2
Personal goods	5.1	134.5	134.6	142.3	141.2	138.8	135.2	135.2	136.23	137.0	137.0	136.6	137.2	137.8	138.5	139.1	139.7	140.4	140.4	140.4	140.4	140.4
Other services	0.3	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.00	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Local by sub-group	**48.6**	141.3	141.2	142.2	147.1	146.7	142.7	143.8	144.17	148.2	152.0	155.1	153.1	155.0	156.0	156.3	161.6	162.7	162.5	162.2	164.4	164.7
A.Food and non-alcoholic beverages	17.5	157.5	154.5	156.5	169.3	159.1	147.7	150.4	150.88	153.6	161.1	166.3	159.8	164.9	168.2	169.9	185.1	188.1	188.1	188.1	194.2	195.2
Food	17.5	157.5	154.5	156.5	169.3	159.1	147.7	150.4	150.88	153.6	161.1	166.3	159.8	164.9	168.2	169.9	185.1	188.1	188.1	188.1	194.2	195.2
Non-alcoholic beverages	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
B.Alcoholic beverages, tobacco and kava	3.6	169.6	174.0	177.7	181.4	181.8	183.4	185.2	187.46	187.5	187.5	191.1	196.0	196.0	196.0	196.0	196.0	196.0	196.0	196.0	196.0	196.0
Alcoholic beverages	0.6	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.25	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2	116.2
Tobacco	1.4	133.9	135.4	140.6	140.6	141.7	141.3	141.3	142.8	142.8	142.8	142.8	143.0	143.0	143.0	143.0	143.0	143.0	143.0	143.0	143.0	143.0
Kava	1.7	219.1	227.4	231.4	239.5	239.5	243.4	247.2	251.0	251.0	251.0	259.0	269.7	269.7	269.7	269.7	269.7	269.7	269.7	269.7	269.7	269.7
C.Clothing and Footwear	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Clothing	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Footwear	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.00	0.0	0.0	0.0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
D. Housing, water, electricity, gas and other fuels	4.9	106.2	112.8	112.8	112.8	110.4	110.4	110.4	110.44	139.8	142.9	130.6	130.6	130.7	128.2	125.3	122.6	122.6	119.9	117.0	117.1	117.2
Maintenance and repair of the dwelling	0.2	159.6	159.6	159.6	159.6	159.6	159.6	159.6	159.60	159.6	159.6	159.6	160.6	161.7	162.7	163.8	164.9	164.9	164.9	164.9	164.9	164.9
Water supply and miscellaneous services relating to the dwelling	1.1	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.00	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

- The Excel-based approach forecasts inflation by individual components. Each category is estimated separately, allowing for tailored methods depending on the nature of the item.
- A few components are estimated using regression models, while others are forecasted with ARIMA models depending on their volatility and data characteristics.

Regression Models: Domestic Food

Dependent Variable: DLOG(CPIFDOM_U)
Method: Least Squares
Date: 08/28/26 Time: 15:03
Sample (adjusted): 2024M02 2026M07
Included observations: 30 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.004302	0.005141	0.836778	0.4117
DLOG(CPIFDOM_U(-1))	0.123295	0.092752	1.329301	0.1974
DLOG(YAM_L)	0.115021	0.021351	5.387110	0.0000
DLOG(TARO_F)	0.078863	0.021520	3.664693	0.0014
DLOG(KUMAR)	0.145800	0.034968	4.169511	0.0004
DLOG(PELE)	0.053735	0.024937	2.154818	0.0424
DLOG(CASS)	0.098334	0.031654	3.106508	0.0051
DLOG(PETR)	0.125545	0.124720	1.006611	0.3251

R-squared	0.852287	Mean dependent var	1.67E-05
Adjusted R-squared	0.805288	S.D. dependent var	0.062263
S.E. of regression	0.027474	Akaike info criterion	-4.127958
Sum squared resid	0.016606	Schwarz criterion	-3.754305
Log likelihood	69.91937	Hannan-Quinn criter.	-4.008423
F-statistic	18.13399	Durbin-Watson stat	1.943986
Prob(F-statistic)	0.000000		

Domestic Food Regression Model

- High-frequency indicators include:
  - ✓ Lag of domestic food inflation index
  - ✓ Yam
  - ✓ Taro
  - ✓ Kumara
  - ✓ Pele
  - ✓ Cassava
  - ✓ Petrol
- The model is applied mainly to nowcast the monthly index when weekly data are available, serving as the starting point for Excel calculations (averaging).
- Incorporating the El Niño crisis is the way forward, representing the next step to strengthen forecasts by capturing climate-driven shocks to food supply.

Regression Models: Electricity prices

Dependent Variable: DLOG(ELEC_D)
Method: Least Squares
Date: 09/02/26 Time: 14:16
Sample (adjusted): 2018M02 2026M07
Included observations: 102 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.000162	0.001844	0.087735	0.9303
DLOG(E_TARI)	0.948081	0.031806	29.80804	0.0000

R-squared	0.898839	Mean dependent var	0.003115
Adjusted R-squared	0.897827	S.D. dependent var	0.058169
S.E. of regression	0.018593	Akaike info criterion	-5.112597
Sum squared resid	0.034572	Schwarz criterion	-5.061127
Log likelihood	262.7425	Hannan-Quinn criter.	-5.091755
F-statistic	888.5190	Durbin-Watson stat	2.954818
Prob(F-statistic)	0.000000		

Domestic Electricity, Gas, and Other Fuels Regression Model

- The quarterly electricity tariff is currently the only explanatory variable used in this model.
- To forecast tariffs, we use ARIMA model with three lags of oil prices as regressors.
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating the index of this group.

Regression Models: Fuel Prices

Dependent Variable: DLOG(OPER_I)
Method: Least Squares
Date: 09/02/26 Time: 16:16
Sample (adjusted): 2018M02 2026M07
Included observations: 102 after adjustments

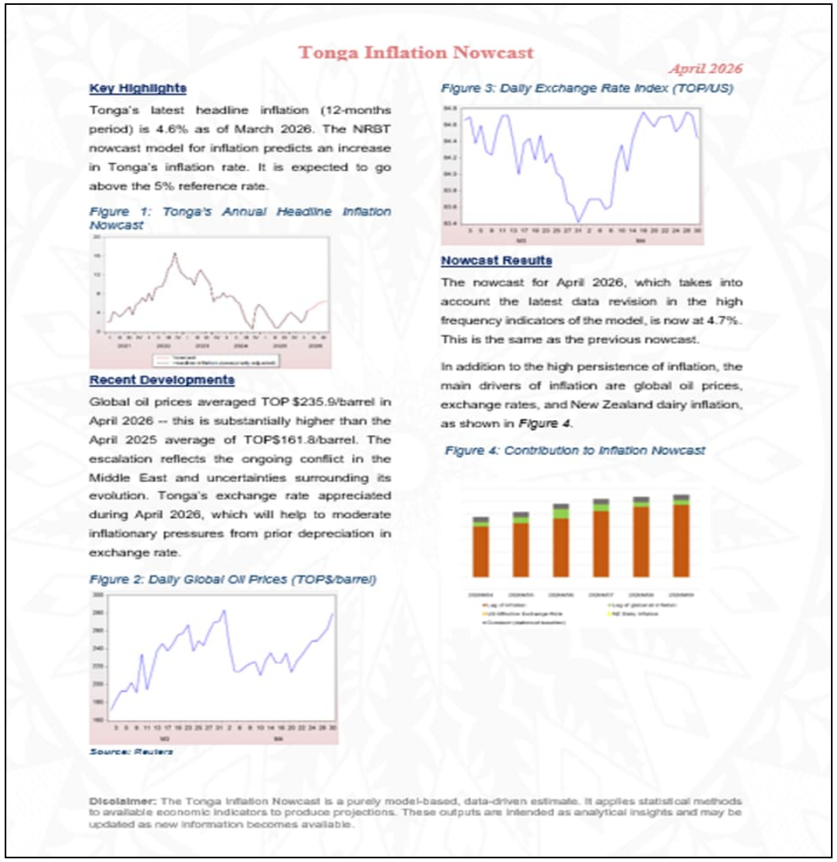
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.002276	0.002038	1.116329	0.2670
DLOG(OIL_P(-1))	0.155385	0.022625	6.867775	0.0000
DLOG(OIL_P(-2))	0.241868	0.024432	9.899691	0.0000
DLOG(OIL_P(-3))	0.042543	0.023187	1.834783	0.0696

R-squared	0.723711	Mean dependent var	0.004171
Adjusted R-squared	0.715253	S.D. dependent var	0.038486
S.E. of regression	0.020537	Akaike info criterion	-4.894780
Sum squared resid	0.041332	Schwarz criterion	-4.791840
Log likelihood	253.6338	Hannan-Quinn criter.	-4.853096
F-statistic	85.56693	Durbin-Watson stat	1.656984
Prob(F-statistic)	0.000000		

Imported Operation of Personal Transport Equipment Regression Model

- The main driver in this category is Fuel.
- High-frequency indicators include:
  - ✓ Three lags of Oil Prices
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating the index of this group.

Reporting Inflation Nowcast



Tonga Inflation Nowcast

April 2026

Key Highlights

Tonga's latest headline inflation (12-months period) is 4.6% as of March 2026. The NRBT nowcast model for inflation predicts an increase in Tonga's inflation rate. It is expected to go above the 5% reference rate.

Figure 1: Tonga's Annual Headline Inflation Nowcast

Recent Developments

Global oil prices averaged TOP $235.9/barrel in April 2026 -- this is substantially higher than the April 2025 average of TOP$161.8/barrel. The escalation reflects the ongoing conflict in the Middle East and uncertainties surrounding its evolution. Tonga's exchange rate appreciated during April 2026, which will help to moderate inflationary pressures from prior depreciation in exchange rate.

Figure 2: Daily Global Oil Prices (TOP$/barrel)

Source: Reuters

Figure 3: Daily Exchange Rate Index (TOP/US)

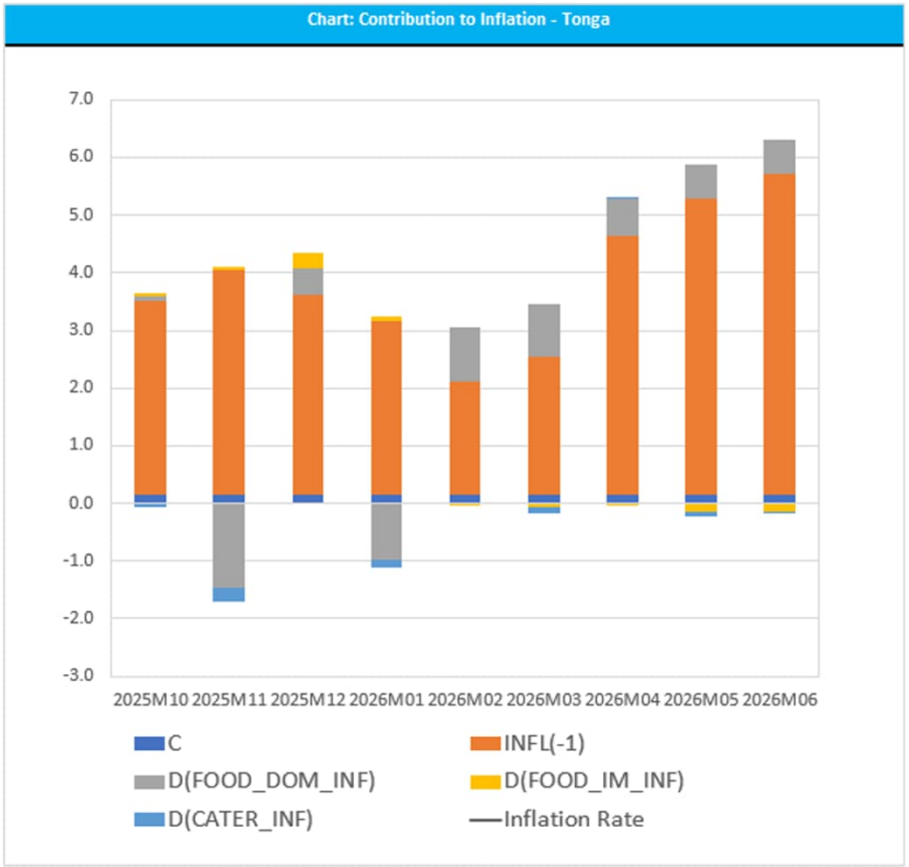
Nowcast Results

The nowcast for April 2026, which takes into account the latest data revision in the high frequency indicators of the model, is now at 4.7%. This is the same as the previous nowcast.

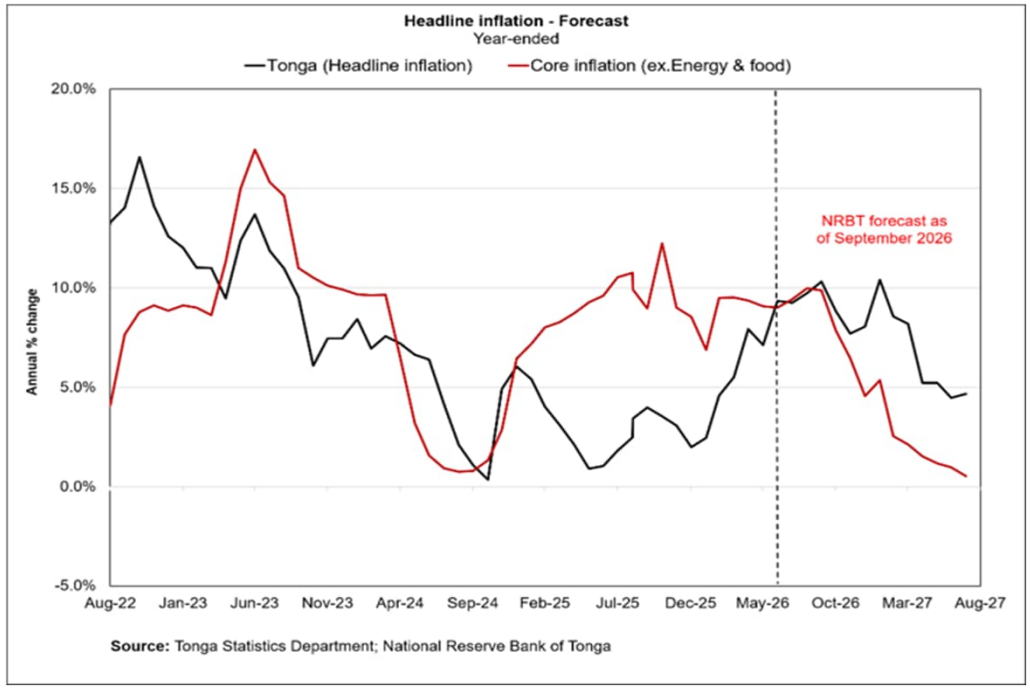
In addition to the high persistence of inflation, the main drivers of inflation are global oil prices, exchange rates, and New Zealand dairy inflation, as shown in ***Figure 4***.

Figure 4: Contribution to Inflation Nowcast

Disclaimer: The Tonga Inflation Nowcast is a purely model-based, data-driven estimate. It applies statistical methods to available economic indicators to produce projections. These outputs are intended as analytical insights and may be updated as new information becomes available.



Inflation Forecast



NRBT Inflation Forecast	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27
Headline inflation (Excel, Regression, ARIMA)	9.2%	9.7%	10.3%	8.8%	7.7%	8.1%
Core inflation	9.4%	10.0%	9.9%	7.9%	6.5%	4.6%
Nowcast Model Average	8.2%	7.6%	7.1%	6.8%	6.6%	6.5%

- The results presented are from the econometric models, including NRBT's estimates from its conventional forecasting approach (i.e., Excel-based).
- We now provide a range of forecasts to Management. Prior to the TA project, only a single, baseline forecast was produced (i.e., Excel-based).

3. GDP Nowcast

Dependent Variable: DLOG(GDP_R)
Method: Least Squares
Date: 07/30/26 Time: 15:59
Sample (adjusted): 2008 2025
Included observations: 18 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.024345	0.008831	2.756782	0.0187
DLOG(GDP_R(-1))	-0.341457	0.253505	-1.346944	0.2051
DLOG(IMP_CONST_R)	0.016756	0.012958	1.293133	0.2225
DLOG(AGR_DOMVAL_R)	0.005599	0.005917	0.946273	0.3643
DLOG(AGR_EXPV_R)	0.026939	0.021803	1.235599	0.2423
DLOG(REM_R)	0.045485	0.054545	0.833908	0.4221
CRISIS	-0.025898	0.016574	-1.562612	0.1464

R-squared	0.435643	Mean dependent var	0.015828
Adjusted R-squared	0.127812	S.D. dependent var	0.029343
S.E. of regression	0.027404	Akaike info criterion	-4.070947
Sum squared resid	0.008261	Schwarz criterion	-3.724691
Log likelihood	43.63852	Hannan-Quinn criter.	-4.023203
F-statistic	1.415202	Durbin-Watson stat	1.879071
Prob(F-statistic)	0.291955		

- This model was recognized as the most effective during the last mission in May 2026.
- The high-frequency indicators include:
  - ✓ Lag of real GDP growth,
  - ✓ Construction imports payments
  - ✓ Domestic agricultural value
  - ✓ Agricultural exports value from the trade report
  - ✓ Remittances
  - ✓ Crisis variable
- The model supports the Excel-based analysis, providing directional insights on GDP growth.
- Further refinement is required, as the Adjusted R-squared remains relatively low.

GDP Forecast: Excel-based

Assumptions	2020/21	2021/22	2022/23	2023/24	2024/25f	2025/26f	2026/27f	2027/28f
	Actual	Actual	Actual	Actual	Actual	Forecast	Forecast	Forecast
2 Sectoral % growth of value-added at 2016/17 constant factor cost						Jul-26	Jul-26	Jul-26
1 Agriculture	-5.3	-1.2	3.2	-10.1	3.57	1.39	-4.00	1.00
2 Fishing	-11.2	-6.5	9.2	-25.0	-2.50	2.00	1.00	1.00
3 Mining and quarrying	10.0	-35.8	-32.2	50.2	4.34	2.00	2.00	2.00
4 Manufacturing	7.3	-21.9	5.4	0.4	2.00	2.00	1.00	2.00
5 Electricity, water and waste	1.3	4.0	6.2	-1.9	8.59	3.00	3.00	2.00
6 Construction	22.1	40.8	-15.5	21.4	6.75	5.00	4.00	3.00
7 Wholesale and retail trade	-0.7	-14.8	5.4	6.3	5.68	2.54	2.00	2.00

- Like Inflation, the Excel-based approach forecasts GDP by individual sector. Each sector is estimated separately, allowing for tailored methods depending on the nature of the sector and data availability.
- Regression models are applied to estimate growth in Agriculture, Construction, Wholesale and Retail Trade, and Accommodation and Food Services.
- We are also exploring the use of ARIMA models to forecast growth in the remaining sectors and incorporate into the excel-based calculations. Further work will be required to identify and explore suitable data series to serve as regressors for these ARIMA-based estimations.

GDP Nowcast: Primary Sector

Dependent Variable: DLOG(AGRI_GDP)
Method: Least Squares
Date: 08/03/26 Time: 16:37
Sample (adjusted): 2001 2025
Included observations: 25 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.007454	0.012389	0.601695	0.5538
DLOG(AGR_EXPV_R)	0.005397	0.024113	0.223830	0.8251
DLOG(AGR_DOMVAL_R)	0.019522	0.011464	1.702898	0.1033
CRISIS_AGR	-0.052194	0.036393	-1.434166	0.1662

R-squared	0.205951	Mean dependent var	0.001044
Adjusted R-squared	0.092516	S.D. dependent var	0.060567
S.E. of regression	0.057697	Akaike info criterion	-2.721579
Sum squared resid	0.069908	Schwarz criterion	-2.526559
Log likelihood	38.01973	Hannan-Quinn criter.	-2.667488
F-statistic	1.815582	Durbin-Watson stat	2.203663
Prob(F-statistic)	0.175220		

Agriculture sector Regression Model

- High-frequency indicators include:
  - ✓ Domestic agricultural value
  - ✓ Agricultural exports value from the trade report
  - ✓ Crisis variable
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating growth for this sector.

GDP Nowcast: Construction

Dependent Variable: DLOG(CONST_GDP)
Method: Least Squares
Date: 07/30/26 Time: 19:11
Sample (adjusted): 2008 2023
Included observations: 16 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.033096	0.049789	-0.664734	0.5188
DLOG(IMP_CONST_R)	0.273069	0.087839	3.108759	0.0090
DLOG(GRANT_REC_R_M...	0.336349	0.153630	2.189346	0.0491
DLOG(REM_R(-1))	0.351810	0.316756	1.110668	0.2885

R-squared	0.500265	Mean dependent var	0.049994
Adjusted R-squared	0.375331	S.D. dependent var	0.216797
S.E. of regression	0.171348	Akaike info criterion	-0.477923
Sum squared resid	0.352322	Schwarz criterion	-0.284775
Log likelihood	7.823381	Hannan-Quinn criter.	-0.468032
F-statistic	4.004236	Durbin-Watson stat	1.917838
Prob(F-statistic)	0.034488		

Construction sector Regression Model

- High-frequency indicators include:
  - ✓ Construction imports payments
  - ✓ Grant receipts
  - ✓ Lag of remittances
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating growth for this sector.

GDP Nowcast: Trading Industry

Dependent Variable: DLOG(W_R_GDP)
Method: Least Squares
Date: 08/03/26 Time: 15:29
Sample (adjusted): 2008 2025
Included observations: 18 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.012830	0.014512	0.884062	0.3916
DLOG(W_R_PAY_R)	0.083336	0.043863	1.899931	0.0782
DLOG(REM_R)	0.127926	0.116068	1.102168	0.2890
DLOG(TRV_REC_R_MOD)	0.040046	0.028423	1.408922	0.1807

R-squared	0.330674	Mean dependent var	0.024696
Adjusted R-squared	0.187247	S.D. dependent var	0.064799
S.E. of regression	0.058418	Akaike info criterion	-2.649250
Sum squared resid	0.047778	Schwarz criterion	-2.451390
Log likelihood	27.84325	Hannan-Quinn criter.	-2.621968
F-statistic	2.305522	Durbin-Watson stat	1.898736
Prob(F-statistic)	0.121278		

Wholesale & Retail Trade sector Regression Model

- High-frequency indicators include:
  - ✓ Wholesale & Retail imports payments
  - ✓ Remittances
  - ✓ Travel receipts
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating growth for this sector.

GDP Nowcast: Tourism Industry

Dependent Variable: DLOG(ACCOMM_GDP)
Method: Least Squares
Date: 07/31/26 Time: 11:35
Sample (adjusted): 1999 2025
Included observations: 27 after adjustments

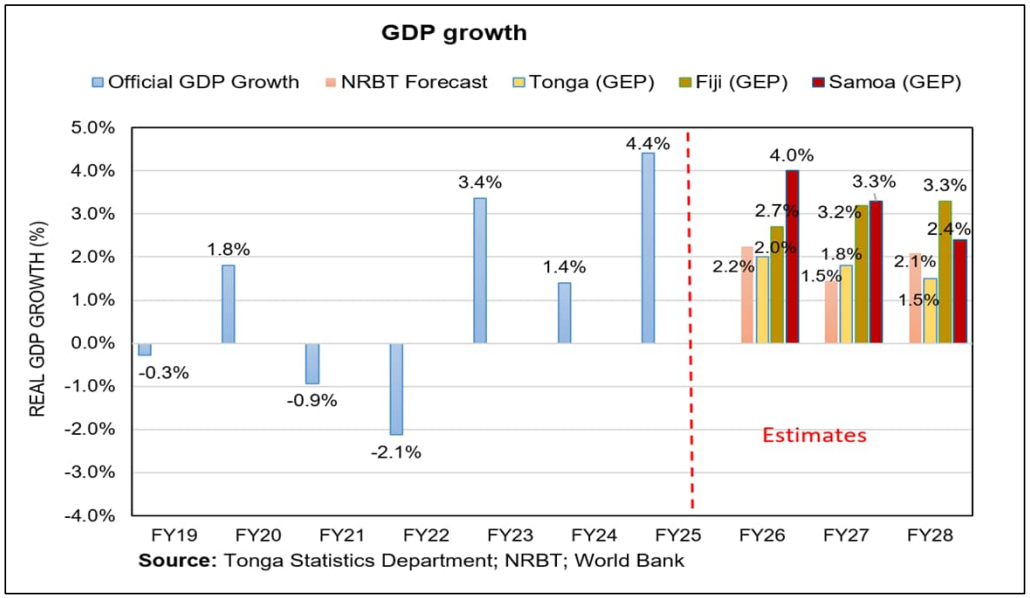
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.003599	0.034901	-0.103133	0.9187
DLOG(REM_R)	0.081581	0.278587	0.292838	0.7722
DLOG(TRV_REC_R_MOD)	0.400085	0.077250	5.179069	0.0000

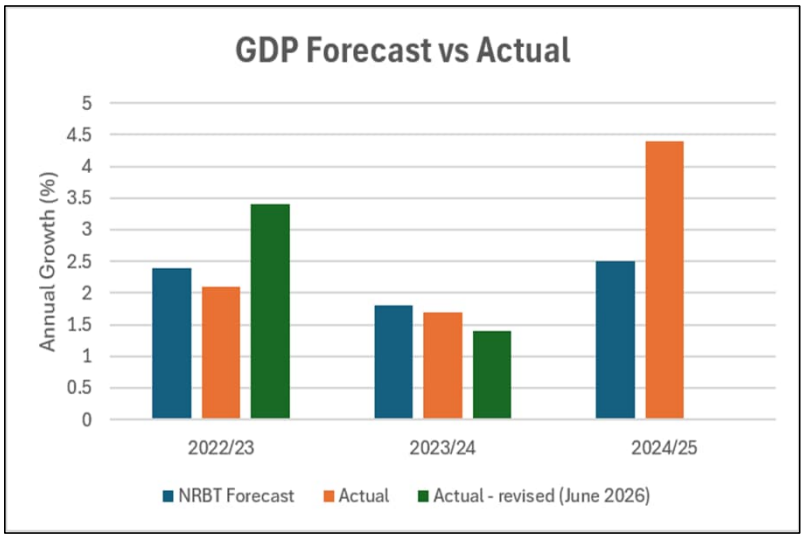
R-squared	0.554331	Mean dependent var	0.015411
Adjusted R-squared	0.517192	S.D. dependent var	0.250321
S.E. of regression	0.173934	Akaike info criterion	-0.555842
Sum squared resid	0.726073	Schwarz criterion	-0.411860
Log likelihood	10.50386	Hannan-Quinn criter.	-0.513028
F-statistic	14.92579	Durbin-Watson stat	1.736673
Prob(F-statistic)	0.000061		

Accommodation and Food Service Regression Model

- High-frequency indicators include:
  - ✓ Remittances
  - ✓ Travel receipts
- To keep it simple for a start, this provides a baseline approach that can be refined over time as more data and methods become available.
- The main purpose of this model is to improve forecasting and develop a more appropriate calculation for estimating growth for this sector.

GDP Forecast: Monetary Policy Statement





NRBT GDP Forecast (%)	2025/26f	2026/27f	2027/28f
Excel-based (NRBT staff estimates, Regression, ARIMA)	2.2	1.5	2.1
Excel-based (Regression, ARIMA)	1.1	1.0	2.1
Nowcast	2.0	1.9	2.0

Bi-annual Monetary Policy Statement

(February and August)

- MPC: Management and Executives
- Board of Directors
- Industry and Stakeholder consultations
- Media Conference and Press release
- Minister MOF, Cabinet, Parliament

4. External Sector: Imports Nowcast

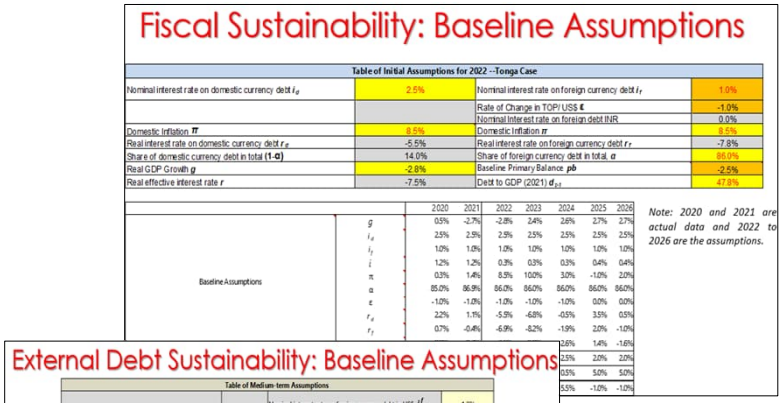
Dependent Variable: DLOG(IMP_D11)
Method: Least Squares
Date: 07/07/26 Time: 14:30
Sample (adjusted): 2007Q2 2025Q4
Included observations: 75 after adjustments

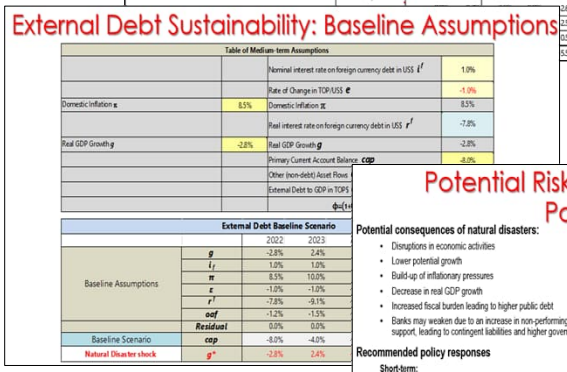
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.019427	0.012647	1.536094	0.1291
DLOG(IMP_D11(-1))	-0.327454	0.112089	-2.921382	0.0047
DLOG(IMP_PAY_JUNE27_D...	0.214063	0.101798	2.102813	0.0391
DLOG(REG_JUNE27_D11)	0.108186	0.083652	1.293289	0.2002
DLOG(RR_JUNE27)	0.140054	0.468682	0.298825	0.7660
CRISIS	-0.036615	0.031096	-1.177467	0.2431

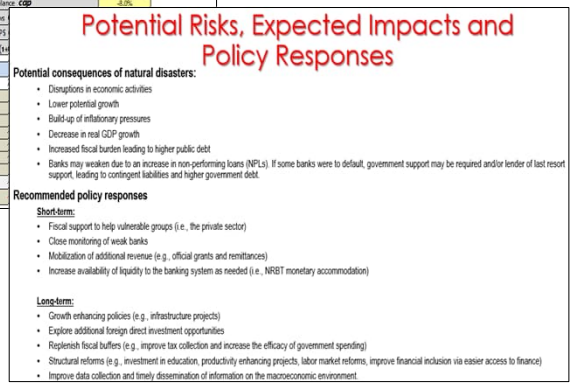
R-squared	0.178994	Mean dependent var	0.013465
Adjusted R-squared	0.119501	S.D. dependent var	0.103634
S.E. of regression	0.097245	Akaike info criterion	-1.746558
Sum squared resid	0.652498	Schwarz criterion	-1.561159
Log likelihood	71.49591	Hannan-Quinn criter.	-1.672530
F-statistic	3.008651	Durbin-Watson stat	2.326787
Prob(F-statistic)	0.016277		

- High-frequency indicators include:
  - ✓ Import payments
  - ✓ Container registrations
  - ✓ REER
  - ✓ Crisis (Covid-19)

5. FPAS: Macroeconomic Diagnostics (MDS)



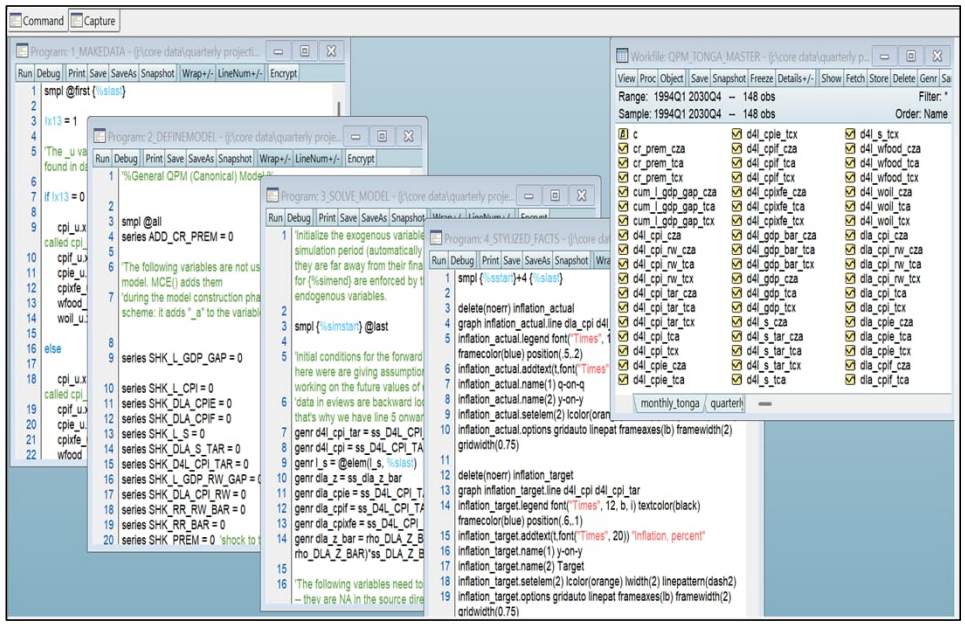




The Macroeconomic Diagnostics is updated every six months in preparation for the Monetary Policy Statement. It provides a comprehensive assessment of the domestic and external economic environments to assist in formulating policy.

- **Real sector assessment**
  - GDP and output gap
  - Business cycle
  - Inflation
- **Financial sector assessment**
  - Credit growth
  - Financial Soundness Indicators
  - Vulnerabilities and Risks
- **Fiscal sector assessment**
  - Government revenue and expenditure
  - Government debt
  - Primary and Overall Balance
  - Fiscal Sustainability
  - Debt to GDP Trajectory
- **External sector assessment**
  - Trade balance
  - Financial account
  - Savings – Investment Gap
  - Foreign Direct Investment
  - Foreign Reserves
  - External debt and sustainability
  - Vulnerabilities and Risks
- **Inter-linkages and Overall Risk Assessment**
  - Potential Risks, Expected Impacts and Policy Responses

6. FPAS: Quarterly Projection Model (QPM)



- The Quarterly Projection Model (QPM) allows for more comprehensive macroeconomic analysis of current and projected economic conditions. It guides management with the decision-making processes by assessing the impact of potential policy changes or shocks to the economy.

- The most critical part of the QPM is the **calibration process**, ensuring the model accurately reflects and aligns with the current economic conditions.

- The entire QPM process – from data importation to forecast generation and the incorporation of policy shocks, is fully scripted within EViews.

- Another fundamental and critical function in this process is the effective management and continuous updating of data.

Canonical Quarterly Projection Model (QPM)

Aggregate Demand Equation:

- $\hat{y}_t = b_1 \hat{y}_{t-1} - b_2 mci_t + b_3 \hat{y}_t^* + \varepsilon_t^y,$

$$mci_t = b_4 \hat{r}_t + (1 - b_4)(-\hat{z}_t)$$

Aggregate Supply Equation:

- $\pi_t = a_1 \pi_{t-1} + (1 - a_1) E\{\pi_{t+1}\} + a_2 rmc_t + \varepsilon_t^{\pi},$

$$rmc_t = a_3 \hat{y}_t + (1 - a_3)\hat{z}_t$$

Equations are interlinked

Colour Match

Uncovered Interest Rate Parity Equation:

- $s_t = E_t\{s_{t+1}\} + \frac{i_t^* - i_t + prem_t}{4} + \varepsilon_t^s$

Taylor Rule

- $i_t = g_1 i_{t-1} + (1 - g_1)(i_t^n + g_2(E_t\{\pi_{t+N}\} - \pi_{t+N}^T) + g_3 \hat{y}_t) + \varepsilon_t^i$

QPM result: Middle East crisis

Alternative vs Baseline: Impacts of the Middle East crisis

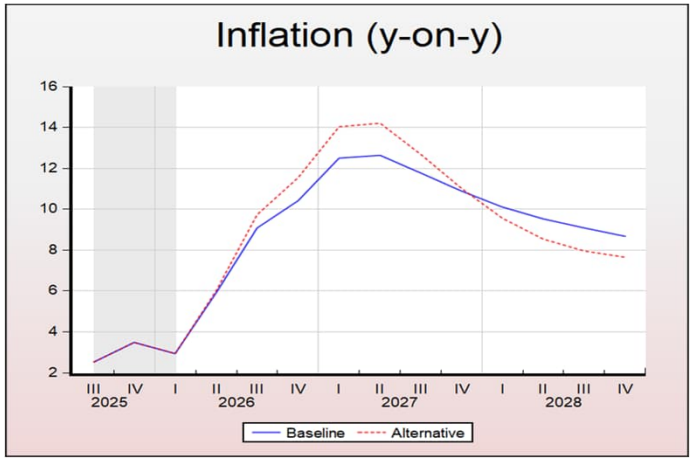


Table 1:	Global Oil Prices Gap in Real Terms	Global Output Gap
2025Q4		
2026Q1		
2026Q2		
2026Q3	10.0%	
2026Q4	5.0%	-0.25%
2027Q1	5.0%	-1.0%
2027Q2		

- To analyze the impact of the Middle East conflict on Tongan economy from the second quarter of 2026 until end of 2028.
- The conflict is modelled here as a shock to global oil prices and the slowdown of the world economy, which presented in Table 1.

Shock 1: Global Oil Prices Gap in Real Terms

- We assume that the Middle East conflict disrupts energy markets, causing oil prices to remain around $100 per-barrel for the next 3 quarters.

Shock 2: Global Output Gap

- The sharp increase in oil prices triggers inflationary pressures and reduces consumer spending worldwide.
- We assume that global GDP growth slows significantly, widening the global output gap.

- In this exercise, the baseline scenario incorporates the actual energy prices for 2026Q1 and 2026Q2. The higher energy prices due to the crisis explain the upward trajectory in the beginning of 2026.
- Following the assumed energy price shock of 10% in 2026Q3 and 5% in 2026Q4-2027Q1, the inflation pathway under this alternative scenario rises above the baseline up until 2027Q4.
- Inflation peaks in 2027Q1 and slows down in 2027Q3 in both scenarios.

7. Key Lessons

i. Effective data management is fundamental for improving efficiency and increasing accuracy.

ii. Search for new data: Tax data, GFS data, Big data, etc.

iii. Refine nowcasting models on a regular basis – constructive dialogue and feedback on the latest economic conditions.

iv. Enhance communication of Nowcast results. – language, visuals, different stakeholders.

v. Capacity Development - continuity.


Events
NRBT